



aisle



All



department



All



24M

Total Product Orders

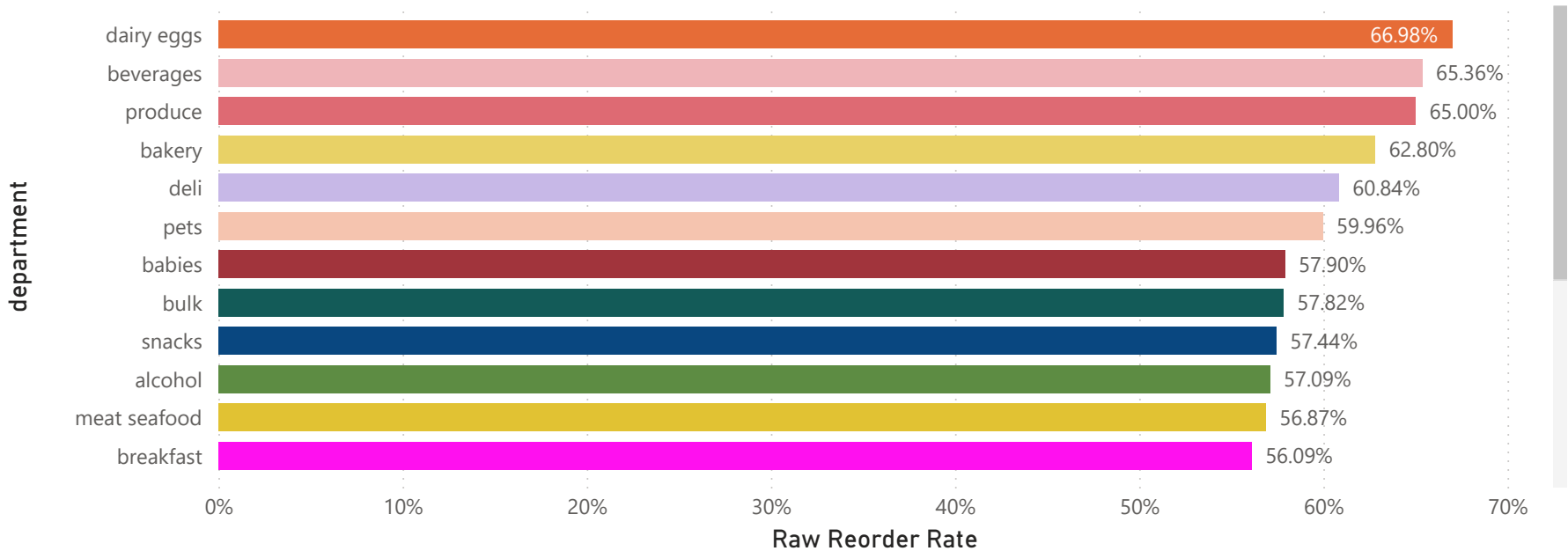
58.98%

Raw Reorder Rate

9689

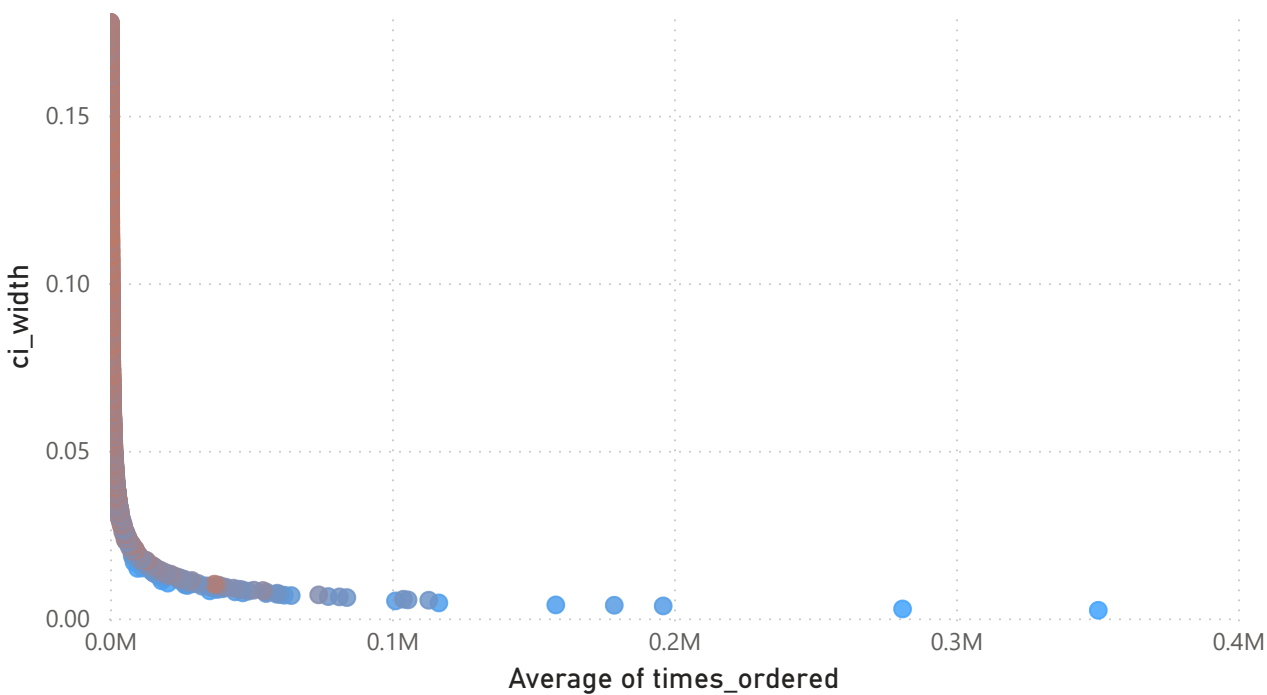
Low-Data Products

Raw Reorder Rate by department

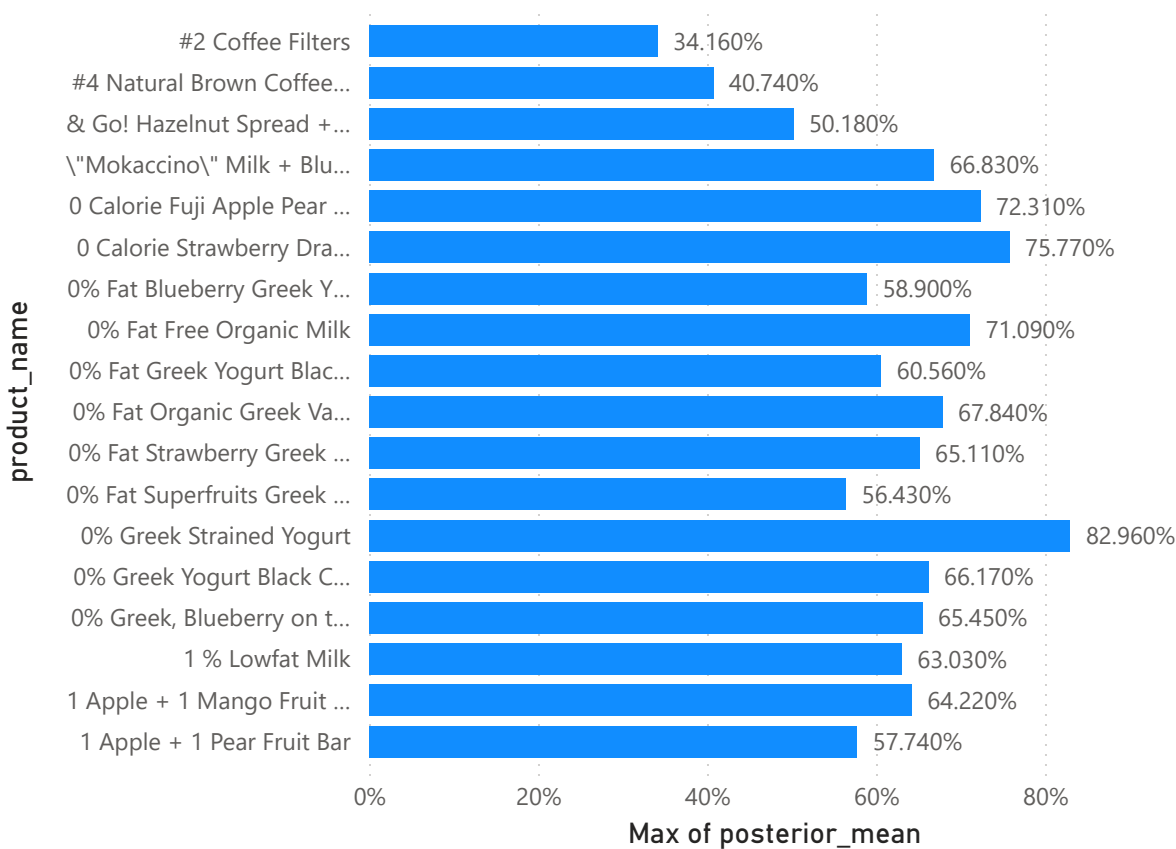


Average of times_ordered by ci_width

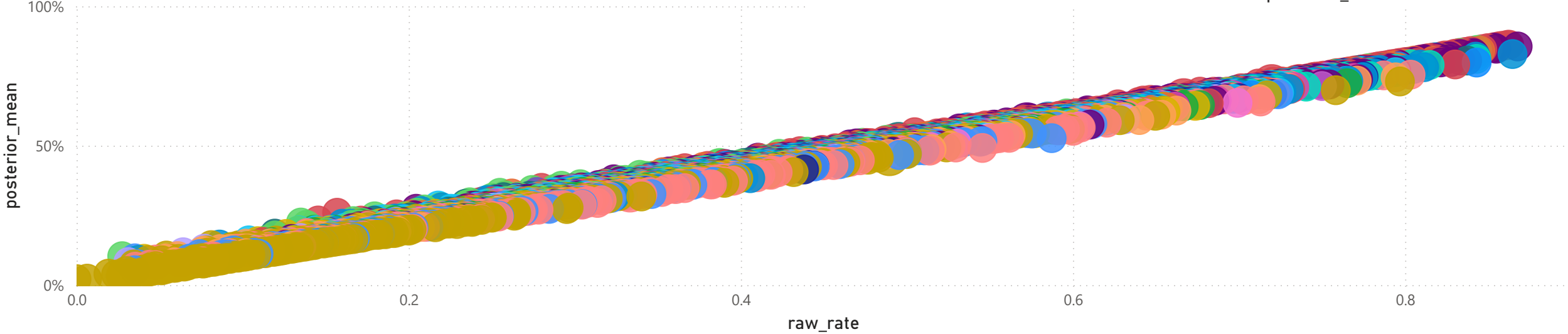
Average of posterior_mean 30.705% 84.340%



Max of posterior_mean by product_name



Count of times_ordered by department, raw_rate and posterior_mean



aisle

All

department

All

product_name	aisle	times_ordered	raw_rate	posterior_mean	ci_lower	ci_upper	ci_width
Half And Half Ultra Pasteurized	milk	2177	0.863	86.190%	0.847	0.876	0.029
Whole Organic Omega 3 Milk	milk	6796	0.862	86.160%	0.853	0.870	0.016
Organic Lactose Free Whole Milk	soy lactosefree	6289	0.859	85.830%	0.850	0.867	0.017
Fragrance Free Clay with Natural Odor Eliminator Cat Litter	cat food care	96	0.906	85.680%	0.788	0.914	0.126
Milk, Organic, Vitamin D	milk	14941	0.857	85.660%	0.851	0.862	0.011
Maca Buttercups	candy chocolate	81	0.926	85.610%	0.782	0.917	0.136
Lo-Carb Energy Drink	energy sports drinks	348	0.868	85.590%	0.818	0.890	0.072
Chocolate Love Bar	candy chocolate	73	0.932	85.450%	0.777	0.918	0.142
Organic Homogenized Whole Milk	milk	2950	0.854	85.310%	0.840	0.866	0.026
Ultra-Purified Water	water seltzer sparkling water	1101	0.854	85.160%	0.830	0.872	0.042
Organic Reduced Fat Milk	milk	26384	0.850	84.990%	0.846	0.854	0.009
Sparkling Water	water seltzer sparkling water	49	0.898	84.920%	0.756	0.923	0.166
Real2 Alkalized Water 500 ml	water seltzer sparkling water	324	0.855	84.770%	0.808	0.884	0.076
Organic Blueberry B Mega	refrigerated	76	0.895	84.670%	0.769	0.911	0.143
Goat Milk	milk	3805	0.847	84.620%	0.835	0.857	0.023
Raw Veggie Wrappers	prepared meals	55	0.927	84.550%	0.756	0.918	0.161
Benchbreak Chardonnay	white wines	86	0.895	84.520%	0.771	0.907	0.136
Organic Whole Milk	milk	7240	0.845	84.430%	0.836	0.853	0.017
Banana	fresh fruits	350453	0.843	84.340%	0.842	0.845	0.002
Wheat Sandwich Bread	bread	782	0.845	84.090%	0.815	0.865	0.051
Organic Lowfat 1% Milk	milk	11065	0.840	83.940%	0.833	0.846	0.014
Organic Reduced Fat Omega-3 Milk	milk	3794	0.836	83.600%	0.824	0.848	0.024
Core Hydration Water	water seltzer sparkling water	37	0.892	83.500%	0.729	0.919	0.189
Organic Low Fat Milk	milk	6265	0.834	83.410%	0.825	0.843	0.018
Organic Milk Reduced Fat, 2% Milkfat	milk	9457	0.834	83.400%	0.826	0.841	0.015
2% Milkfat Greek Yogurt with Mango On The Bottom	yogurt	72	0.875	83.390%	0.752	0.902	0.151
Bag of Organic Bananas	fresh fruits	281052	0.833	83.300%	0.832	0.834	0.003
Purified Water, 9.5pH+	water seltzer sparkling water	233	0.841	83.240%	0.784	0.876	0.092
Lowfat Goat Milk	milk	869	0.833	83.200%	0.807	0.856	0.049
Purified Alkalkine Water with Minerals pH10	water seltzer sparkling water	545	0.835	83.110%	0.799	0.861	0.062
1% Milkfat Low Fat Vitamin A & D Milk	milk	1089	0.832	83.100%	0.808	0.853	0.044

9689
Low-Data Products

Top Reorder Department
dairy eggs

Strategic Action Plan

1. **Inventory Prioritization:** Focus fulfillment optimization and stock availability on top-performing reorder categories identified by the posterior mean.
2. **Targeted Promotions:** Use Bayesian scores to identify underperforming items with high true potential, optimizing promotional spend without over-investing in volatile low-volume products.

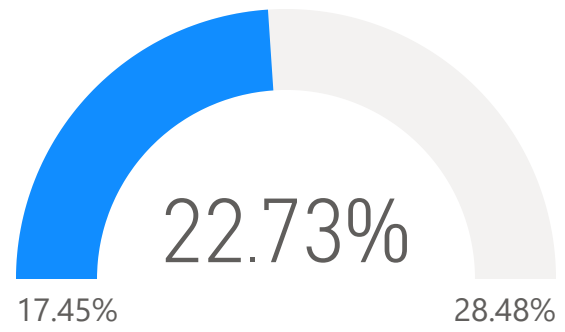
Key Insights & Methodology

- **Raw Rate vs. Bayesian Estimation:** Standard reorder rates create extreme, noisy estimates for low-volume products. Empirical Bayes posterior probabilities smooth risk-adjusted metrics to prevent anomalies from skewing rankings.
- **Top Department Stability:** High-frequency categories (Dairy & Eggs, Produce) exhibit consistent baseline reorder behavior across raw and posterior metrics.
- **Low-Data Volume Adjustment:** Products with limited order histories are pulled toward the global mean rate, ensuring decisions rely on statistically reliable signals.

← Estimated Discrepancy Rate

22.73%

95% High-Density Interval
(HDI) Confidence Range



MODERATE_MISCOUNT_RISK

First stock_risk_category

stock_risk_category	posterior_mean	hdi_95_lower	hdi_95_upper	Sum of posterior_alpha	Sum of posterior_beta
MODERATE_MISCOUNT_RISK	22.73%	17.45%	28.48%	50.00	170.00
Total				50.00	170.00